

COLLEGE OS

AI-Powered Digital Campus Platform

PROJECT PLANNING & SOURCE OF TRUTH BLUEPRINT

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PROGRAM & SEMESTER

B.Tech (CSE) • 7th Semester

PROJECT GUIDE

[Project Guide Name]

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DOCUMENT VERSION & STATUS

Version 1.1 • Foundation Updated — Auth & Onboarding Locked

 SOURCE OF TRUTH POLICY & GOVERNANCE DIRECTIVE

VERSION 1.1 LOCKED

Purpose & Authority: This document establishes the authoritative, finalized planning decisions for **College OS**. All sections marked with  **LOCKED — SOURCE OF TRUTH** are permanently binding across all development phases, backend architectures, MongoDB schemas, Next.js frontend modules, and Campus AI retrieval pipelines.

Version 1.1 Revision Changelog (Foundation Updated — Authentication & Institutional Onboarding Locked):

- Added institution-controlled identity architecture ("Users do not choose their institutional role; the institution provisions/authorizes it").
 - Replaced unrestricted role-based signup concept; eliminated public self-registration across all roles.
 - Locked student account activation workflow via verified institutional records (Enrollment Number + Official Contact + OTP).
 - Locked faculty/staff activation workflow (Employee ID + Official Contact + OTP).
 - Clarified HOD / Principal / VP provisioning (institution-provisioned; authenticated via Staff/Institutional Login; zero self-elevation).
 - Added account lifecycle states: INVITED / NOT_ACTIVATED , ACTIVE , SUSPENDED , GRADUATED / INACTIVE .
 - Clarified strict separation of institution-controlled academic identity data vs. user-controlled showcase profile data.
 - Updated authentication UI model: two clean entry paths (Student Portal vs. Faculty & College Staff Portal).
 - Updated MongoDB operational schemas (users , studentProfiles , facultyProfiles), RBAC matrices, and Campus AI authority pipelines.
- **Authoritative Rule:** Engineering execution must strictly implement these locked decisions without arbitrary deviations.
 - **Change Control:** Any modification requires explicit justification against the 4-Question Evaluation Gate and formal increment of document versioning.
 - **Scope Discipline:** Adheres rigorously to *Core First → Stabilize → Expand*. MVP delivery shall not be delayed by experimental ideas.

DOCUMENT OUTLINE • TABLE OF CONTENTS

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STEP 1 • PROJECT FOUNDATION & SCOPE

Source of Truth — Version 1.1 • Baseline Architectural Planning — Authentication & Institutional Onboarding Locked

DOCUMENT PURPOSE & SCOPE BOUNDARY

This section defines the fundamental identity, problem landscape, core objectives, user categorizations, scope boundaries, and functional modules of the College OS project. All downstream engineering artifacts—user journeys, RBAC permission tables, database schemas, API routes, user interfaces, AI pipelines, and roadmap milestones—are strictly derived from this baseline.

1. Project Identity LOCKED

1.1 Working Project Name

College OS

1.2 Naming Status

Working Name — NOT FINAL: The permanent product or brand name will be selected prior to public launch. For all current architectural planning, system design, academic documentation, and repository tracking, the name **College OS** is the authoritative working designation.

2. Project Vision LOCKED

College OS is engineered as a **centralized digital campus ecosystem** that unifies academic management, institutional communication, student identity, peer discovery, and context-aware intelligence into a single modern platform.

Crucially, College OS is **not intended to be just another college ERP**. While traditional academic ERPs treat students strictly as records in an administrative database, College OS builds a holistic digital environment where:

- Students manage and monitor their real-time academic status (attendance, timetables, marks, assignments).
- Students build and maintain a professional, verified campus identity and portfolio.
- Students discover peers, explore student-built projects, and collaborate across departments.
- Faculty members manage their assigned classes and evaluate academics within strictly bounded permissions.
- The institution maintains an authentic, centralized digital presence for official communications.
- All critical campus information is accessible from a unified, single source of truth.
- Students interact with campus resources through an AI-powered, context-aware Campus Assistant.
- Future campus utilities (internships, hackathons, lost & found) can be integrated modularly without fragmenting into external apps.

CORE VISION STATEMENT

"One connected digital campus for academics, institutional information, student identity, interaction and intelligent information access."

3. Problem Statement & Campus Fragmentation LOCKED

3.1 The Primary Fragmentation Problem

In contemporary higher education ecosystems, academic data, institutional updates, and student activities are distributed across chaotic, uncoordinated channels. A student is forced to navigate:

CHANNEL	INFORMATION HANDLED	CRITICAL FAILURE POINT
College ERP / MyCamu	Attendance records, basic fee status, exam registration	Rigid, clunky UX; lacks modern student-facing identity and collaboration tools
WhatsApp Groups	Assignments, unofficial notices, schedules, links	No searchability, spam-heavy, unverified messages, zero data persistence
Physical Notice Boards / PDFs	Circulars, timetable changes, exam timetables	Outdated quickly, easily missed, zero notification mechanics
College Web Portals	General institutional information, admissions	Static, disconnected from daily student academic activities
Faculty Direct Messages	Subject assignments, submission guidelines, deadlines	Scattered across personal chat apps with zero central tracking
External Job & Event Portals	Internships, hackathons, competitions	Generic platforms lacking college relevance or peer team-formation context

3.2 Academic Information Fragmentation

Core academic assets—daily schedules, real-time attendance percentages, evaluation marks, assignments, and exam dates—exist in disparate silos. Students have no single, reliable dashboard to answer: *"Where do I need to be next? What assignment is due? What is my current attendance buffer?"*

3.3 Limited Student Identity & Discoverability Deficit

Conventional college systems treat students solely as administrative rows (Roll Number, Fee Status). There is no professional platform for students to showcase verified technical skills, GitHub profiles, personal websites, and technical projects. Simultaneously, students have no mechanism to discover what their peers are building, identify collaborators, or review past student work. This creates a severe structural gap between **academic compliance** and **campus community innovation**.

4. Existing ERP & MyCamu Gap Analysis **LOCKED**

It is vital to establish that this project does **not** define its objective as "MyCamu is bad."

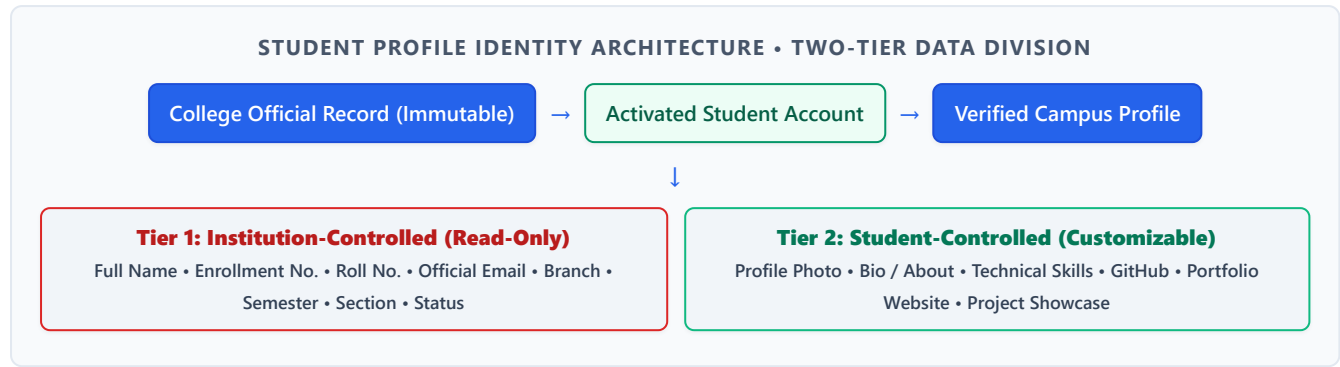
ARCHITECTURAL POSITIONING: EXTENSION, NOT MERE REPLACEMENT

Existing ERP systems effectively solve a significant segment of institutional record-keeping and administrative management. However, they fail to provide a complete student-centric campus platform that bridges academic administration with student portfolio identity, project exploration, institutional broadcasting, and natural-language AI assistance.

Therefore, **College OS is not simply an ERP replacement**; it expands the boundaries of campus software to encompass the full student lifecycle.

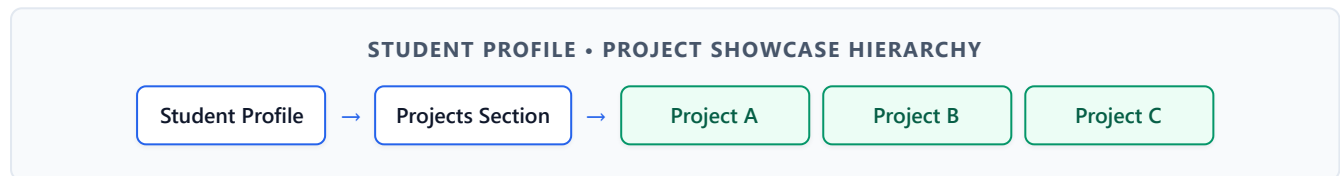
5. Student Interaction & Campus Profile Architecture **LOCKED**

College OS establishes a dedicated student identity model that lives natively within the institutional software environment. A foundational architectural principle separates **institution-controlled academic identity** from **user-controlled profile content**:



6. Student Project Discovery & Interaction **LOCKED**

Project showcasing is a locked, core pillar of College OS. Students can publish projects directly to their profiles, detailing technical specifications, live URLs, code repositories, and media previews.



The platform incorporates a centralized **Explore Students / Student Projects** directory where campus members can discover other students, view profiles, inspect project code repositories, and interact with projects via **Project Likes**.

LOCKED SCOPE BOUNDARY REGARDING SOCIAL INTERACTIONS

The detailed social interaction paradigm (e.g., bidirectional follower graphs, direct real-time messaging, public comments) is **intentionally NOT finalized in Step 1**. The locked core direction is strictly: **Student Profiles + Project Showcasing + Student/Project Discovery + Project Likes**.

7. Opportunity Discovery Gap **PHASE 2**

Students miss critical career accelerators (internships, hackathons, research grants, competitions) because listings are scattered across private WhatsApp chats and external boards. College OS will centralize curated campus opportunities in Phase 2.

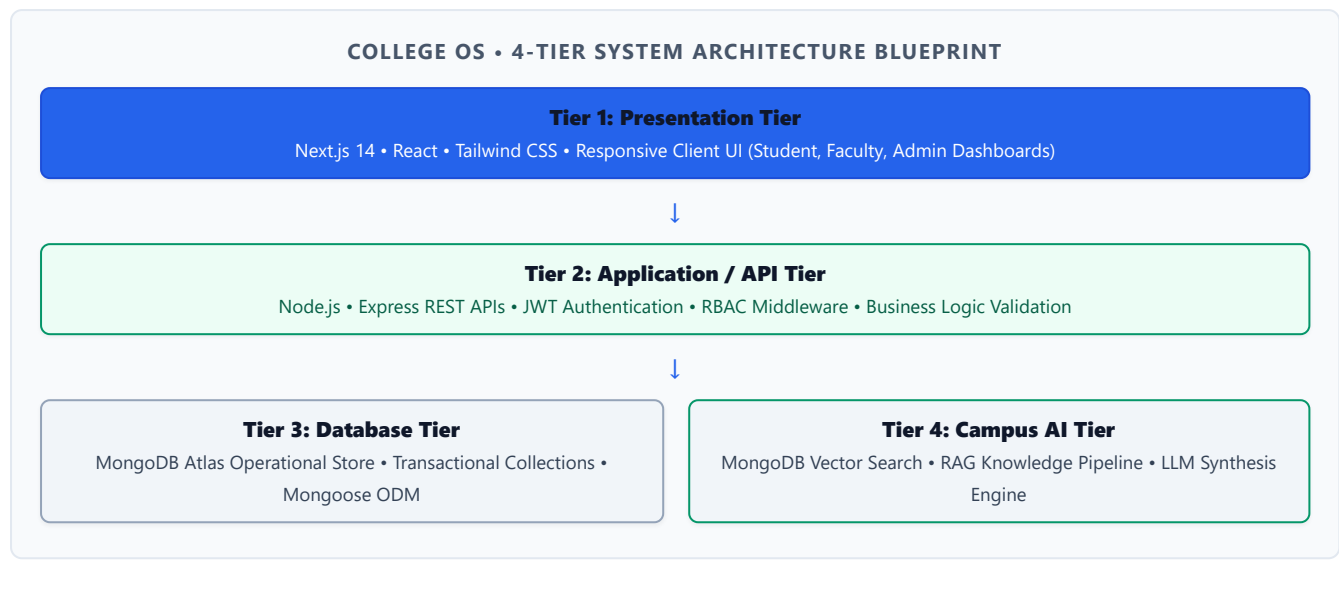
8. Campus Utility Workflows **PHASE 2**

Everyday campus operational friction lacks digital structure. For example, **Lost & Found** currently relies on broadcast spam in classroom WhatsApp groups. College OS plans a dedicated, verified Lost & Found tracking workflow in Phase 2.

9. Proposed Solution • High-Level Architecture **LOCKED**

HIGH-LEVEL SOLUTION DEFINITION

"College OS will provide a centralized digital campus platform that combines academic information, student identity, student/project discovery, institutional communication and AI-assisted access to college-specific information, bounded by strict role-based authority."



10. Core User Types & Institutional Onboarding Architecture **LOCKED V1.1**

CORE INSTITUTIONAL IDENTITY PRINCIPLE

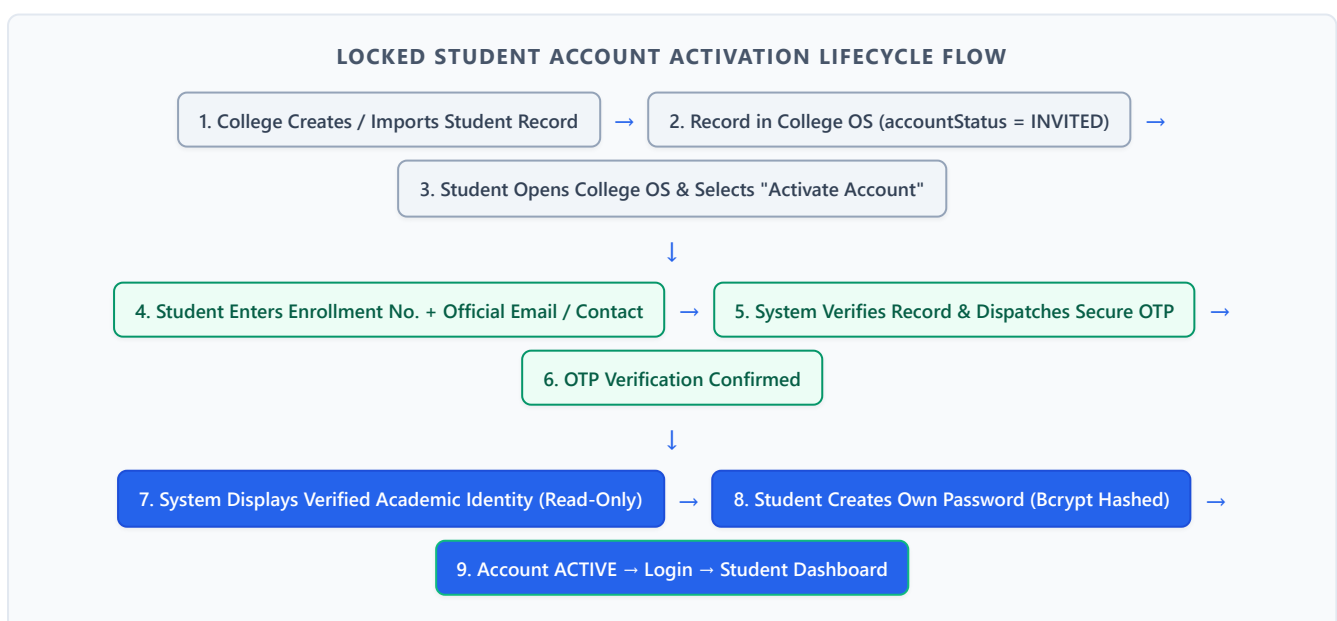
"Users do not choose their institutional role. The institution provisions and authorizes the user's role."

College OS is an institution-controlled digital campus platform, NOT an open public registration portal. A user cannot self-register and arbitrarily select *Student*, *Faculty*, *HOD*, *Principal*, or *VP*. All institutional identities and roles must originate from official college records.

USER ROLE	PROVISIONING & ONBOARDING MECHANISM	FUNCTIONAL SCOPE & ACCESS BOUNDARIES
Student	Institutional Pre-Provisioning → Self-Activation: College imports/creates official record. Student activates account via verified Enrollment No. + Official Email/Phone + OTP, then sets own password.	Primary platform users; restricted strictly to own academic records (attendance, marks, timetable, assignments) and public campus directories (profile showcase, explore directory, project likes, notices, events, Campus AI).
Faculty	Institutional Pre-Provisioning → Self-Activation: College administration provisions faculty record with Employee ID, department, and teaching assignments. Faculty activates via Employee ID + Contact + OTP, then sets password.	Academic managers; granted elevated, strictly scoped permissions bounded dynamically to admin-assigned classes and subjects (attendance marking, assignment creation/grading, marks entry, class notices).
HOD (Head of Department)	Administrative Provisioning Only: Provisioned directly by executive authority with departmental scope. Authenticates via Staff Login. Never self-selectable.	Department-level academic oversight; review faculty class assignments, inspect departmental attendance and grades, publish department bulletins, monitor curricula.
Principal / VP / Dean	Administrative Provisioning Only: Executive-level credentials provisioned by university governance. Authenticates via Staff Login. Never self-selectable.	Campus-wide institutional oversight; campus-wide bulletins, cross-department academic audits, university event authoring, executive reports.
College / Institution Entity	Singleton Institutional Presence: Official profile representing the university/college as an institution.	Official institution profile, verified campus circulars, accredited announcements, master academic calendars (managed by authorized administrative personnel).

10.1 Student Authentication & Account Activation Flow 🔒 LOCKED FLOW

A student does **not** independently invent their institutional identity during signup. Before an account can be activated, the college must already have an official student record in College OS containing: *Full Name, Enrollment Number / Student ID, Official College Email, Phone Number, Department / Branch, Semester, Section, Roll Number, and Academic Status*.



🔒 Student Account Security Rule

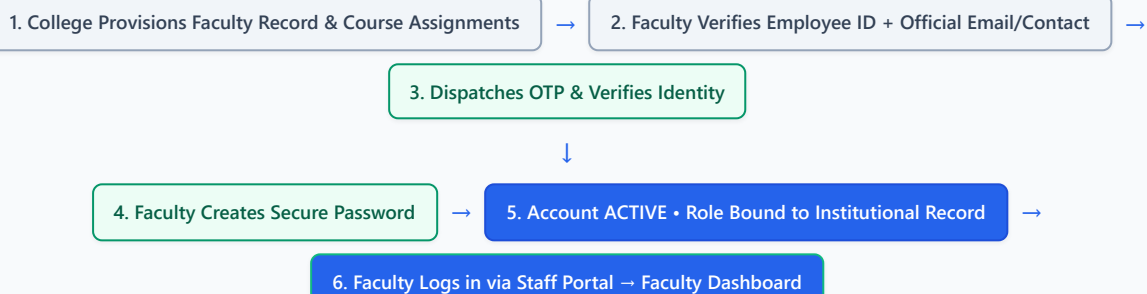
Enrollment Number alone is strictly INSUFFICIENT for activation. To prevent malicious takeover by anyone knowing a student's public enrollment ID, the system requires a multi-factor handshake: `Enrollment Number` + `Official College Email / Verified Phone` + `Time-bound OTP`.

🔒 Password Ownership & Privacy Policy

The college does NOT provide or maintain permanent student passwords. Students create their own passwords during activation. Credentials are cryptographically hashed using `bcrypt` (cost factor ≥ 10). College staff have zero access to plaintext passwords.

10.2 Faculty & Staff Authentication Flow 🔒 LOCKED FLOW

Faculty members follow the identical institution-controlled provisioning principle. The college administration provisions the faculty record (*Employee ID, Full Name, Official Email, Department, Designation, Assigned Classes, Assigned Subjects*).

FACULTY & STAFF INSTITUTIONAL ONBOARDING & ACTIVATION FLOW**10.3 HOD, Principal & Executive Provisioning** 🔒 LOCKED RULE

Zero Public Self-Registration for Institutional Authorities: There is NO public "Sign Up as HOD", "Sign Up as Principal", or "Sign Up as VP". These high-privilege administrative accounts are provisioned exclusively through authorized institutional channels. A user can never elevate their own privileges through user-inputted registration fields.

10.4 Authentication UI Information Architecture 🔒 LOCKED UX

The public-facing authentication interface eliminates all generic role-selection dropdowns. Instead, College OS provides two streamlined authentication paths:

COLLEGE OS AUTHENTICATION PORTAL

- └─ 1. Student Portal
 - └─ Student Login (Email/Enrollment No. + Password)
 - └─ Activate Student Account ("Activate Your College OS Account" • OTP Verification)
- └─ 2. Faculty & College Staff Portal
 - └─ Staff Login (Email/Employee ID + Password • Backend Authoritative Role Resolution)

Note: For staff, the backend securely determines whether the authenticated user is Faculty, HOD, Principal, or VP based strictly on their institutionally provisioned database record.

10.5 Account Status Lifecycle Management **LOCKED STATES**

LIFECYCLE STATE	SYSTEM TRIGGER	ACCESS & OPERATIONAL PERMISSIONS
INVITED / NOT_ACTIVATED	Official record created/imported by college	Pre-provisioned record exists in operational store; cannot log in; pending identity verification and password creation.
ACTIVE	Successful OTP verification + password creation	Full role-permitted access to student or staff dashboard, academic tools, directory, and Campus AI.
SUSPENDED	Administrative disciplinary action or security hold	Authentication blocked; session tokens invalidated; read/write access revoked pending institutional review.
GRADUATED / INACTIVE	Student graduation or faculty offboarding	Archived academic record; read-only portfolio preservation if alumni policy permits; write access revoked.

10.6 Institution-Controlled Data vs. User-Controlled Profile Data **LOCKED BOUNDARY**

DATA CLASSIFICATION	CONTAINED ATTRIBUTES	GOVERNANCE & MUTATION RULES
Institution-Controlled Data <i>Official Academic Identity</i>	Full Name, Enrollment Number / Student ID, Roll Number, Official College Email, Institutional Phone, Department / Branch, Current Semester, Section, Academic Standing, Faculty Employee ID, Faculty Designation, Assigned Classes & Subjects, Institutional Role.	Read-Only for User: Established exclusively via institutional provisioning. The user CANNOT edit, redefine, or forge these values. Updates require administrative verification.
User-Controlled Profile Data <i>Student Portfolio & Showcase</i>	Profile Avatar / Photo, About Me / Bio, Technical Skills tags, GitHub Profile URL, Personal Website / Portfolio Link, Project Showcase submissions, Approved interest tags.	Customizable by User: Editable directly by the authenticated user to build a professional campus presence and project portfolio.

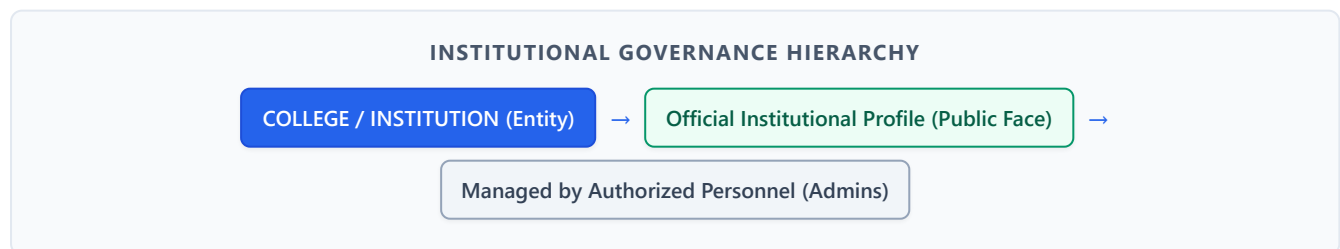
11. College Institutional Identity Architecture **LOCKED**

College OS provides an official **College Profile** entity. This profile represents the institution itself rather than an individual person. It functions as the authenticated campus broadcast station:

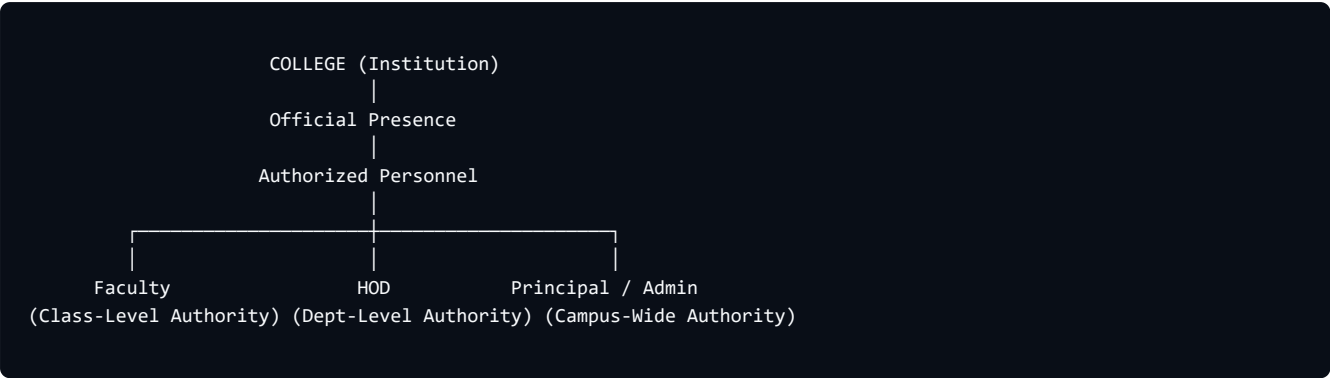
- Official institutional identity, accreditation, and campus details.
- Verified institutional circulars and mandatory academic notices.
- Official college events, annual symposiums, and administrative updates.

12. College Profile vs. Admin Account Separation **LOCKED**

A vital architectural distinction is enforced between the **Institution** and **Administrative Personnel**:



13. Role & Governance Hierarchy LOCKED



14. Core Functionality Specification (The 11 MVP Modules) LOCKED

#	MODULE NAME	FUNCTIONAL SCOPE & RESPONSIBILITIES	PRIMARY USERS
1	Authentication & RBAC	Institution-controlled user provisioning, student account activation (Enrollment + Contact + OTP), faculty/staff activation, role provisioning, secure credential creation, login, session lifecycle (JWT/HttpOnly), RBAC authorization, account lifecycle management, and verification.	All Users
2	Student Profile	Personalized campus profile displaying bio, verified skills, GitHub repository links, personal website, and student metadata.	Students, Peers
3	Student Projects	Showcase technical projects with titles, markdown descriptions, tech stacks, GitHub URLs, live deployment links, and image screenshots.	Students, Faculty
4	Explore Directory	Searchable directory to discover student profiles and peer projects; includes project liking and engagement analytics.	Students, Faculty
5	Timetable Management	Digital schedules for students; customized daily lecture timetables for faculty mapped strictly to assigned subjects.	Students, Faculty
6	Attendance Tracking	Real-time attendance percentage tracking for students; dynamic roll-call marking interface for authorized faculty.	Students, Faculty
7	Marks & Performance	Semester-wise marks, assessment breakdown, and GPA views for students; secure grade entry workflows for faculty.	Students, Faculty
8	Assignments Workflow	Assignment publication, deadline tracking, and file submissions; faculty assignment creation, updates, and grading.	Students, Faculty
9	Notices & Bulletins	Centralized broadcast hub for college-wide, departmental, and class-level notices with priority flags and attachment downloads.	Campus Community
10	Campus Events	Directory of technical symposiums, workshops, seminars, and cultural activities with registration links and scheduling details.	Campus Community
11	Campus AI Assistant	Intelligent natural-language conversational assistant answering queries about attendance, timetables, assignments, and college rules.	Students, Faculty

15. Campus AI Principles & Anti-Hallucination Grounding

🔒 LOCKED

The Campus AI Assistant is **not an unrestricted, generic chatbot** (such as an open ChatGPT wrapper). It is engineered as a domain-bounded, context-grounded college intelligence system:

- **College OS is the Primary Source of Truth:** The assistant's reasoning is anchored strictly in authoritative College OS databases (MongoDB operational store) and verified institutional documents (RAG vector store).
- **One Common Engine with Role-Aware Context & Authoritative RBAC:** Student AI and Faculty AI share ONE common Campus AI engine. The AI authorization pipeline is strictly bounded: *Authenticated User* → *Institution Check* → *Authoritative Backend Role* → *RBAC Permissions* → *Allowed Context* → *Grounded Query*. User-claimed roles from the frontend are never trusted; the authenticated institutional role from the database is authoritative. Queries enforce least privilege, and passwords, credentials, OTPs, session tokens, and unauthorized student/faculty records are strictly excluded.
- **Absolute Anti-Hallucination Mandate:** If an authenticated student or faculty member asks a college question not found in verified records, the assistant explicitly responds: *"This information is currently unavailable in official college records"*, rather than speculating.
- **Contextual Privacy, Least-Privilege & Credential Exclusion:** Queries strictly enforce RBAC (students: own records; faculty: assigned classes/students). The AI never exposes unauthorized peer data, another faculty's records, or credentials.

16. Optional / Phase 2 Features (Planned Expansion)

PHASE 2 PLANNED

To guarantee on-time completion and architectural stability of the core MVP, the following four modules are classified as **Phase 2**:

PHASE 2 EXPANSION ROADMAP

└─

1. Internships Ecosystem (Curated listings, application tracking, student referrals)

└─

2. Hackathons & Competitions (Team formation, event discovery, project submissions)

└─

3. Campus Community Feed (Campus-wide discussions, announcements, peer interactions)

└─

4. Lost & Found Module (Item reporting, verification workflow, resolution tracking)

17. Problem-to-Solution Mapping Matrix

🔒 LOCKED

IDENTIFIED CAMPUS FRICTION	PLANNED MODULE	LIFECYCLE PHASE & RATIONALE
Scattered internship notifications across WhatsApp groups	Internships Module	PHASE 2 • Extends student identity directly into hiring channels.
Scattered hackathon announcements; difficulty finding team members	Hackathons Module	PHASE 2 • Leverages student skills directory for team matching.
Lack of open, moderated campus discussion forums	Campus Community Feed	PHASE 2 • Provides an authenticated campus-wide public square.
Chaotic lost item recovery relying on WhatsApp spam and word-of-mouth	Lost & Found	PHASE 2 • Structured claim, verification, and audit trail.

18. Core MVP vs. Phase 2 Scope Boundaries

🔒 LOCKED

THE FUNDAMENTAL SCOPE BOUNDARY RULE

Build Core First → Stabilize & Test → Expand to Phase 2.

Under no circumstances shall Phase 2 modules be developed prematurely if they distract from or delay the delivery of the 11 Core MVP modules.

Core MVP Scope (Current Focus)

- Authentication, Institutional Identity & RBAC
- Student Profile & Showcase
- Student Projects Showcase
- Explore Students & Projects
- Project Likes System
- Academic Timetable
- Attendance Management
- Marks & Academic Grades
- Assignments Lifecycle
- Notices & Announcements
- Campus Events Directory
- Campus AI Assistant
- Official College Profile

Phase 2 Scope (Subsequent Expansion)

- Curated Internships Portal
- Hackathon & Team Matching
- Campus Community Social Feed
- Digital Lost & Found Workflow
- Direct Student Messaging
- Complex Social Followers Model
- Automated Fee Payment Gateway
- Library Physical Book Barcode Scan

19. What College OS Is NOT (Explicit Non-Goals) 🔒 LOCKED

To protect engineering resources and prevent fatal scope creep, College OS explicitly defines what it is **NOT**:

- **NOT an unrestricted public registration platform:** College OS is not an open social platform where anyone can create a fake institutional identity or self-assign an academic/administrative role. The platform's institutional identity layer must remain strictly college-controlled.
- **NOT a complete 100% replacement for institutional legacy systems:** It integrates with or sits alongside complex backend administration without attempting to rebuild financial accounting engines.
- **NOT a generic, open social network:** It is not Instagram, Facebook, or Twitter; all interactions are academically and professionally focused.
- **NOT a generic ChatGPT clone:** The AI is an anchored institutional tool, not an open-ended conversational gimmick.
- **NOT a commercial job portal:** It does not aim to replace LinkedIn, Indeed, or Naukri.com.
- **NOT an e-commerce platform:** It does not facilitate commercial buying/selling.
- **NOT a real-time messaging-first chat app:** It does not attempt to replicate WhatsApp or Telegram instant messaging in the MVP.
- **NOT an unmanageable collection of every possible campus utility:** Features without rigorous justification are rejected.

20. Project Scope Philosophy 🔒 LOCKED**"CORE FIRST, EXPANSION LATER" EXECUTION LIFECYCLE**

COLLEGE OS PROJECT INITIATIVE →

PHASE 1: CORE MVP (11 Modules) →

Testing & Stabilization →

PHASE 2: EXPANSION

21. Master Product Taxonomy Tree 🔒 LOCKED

COLLEGE OS PLATFORM

- 1. AUTHENTICATION & INSTITUTIONAL IDENTITY
 - Institutional User Provisioning (College Pre-Created Records)
 - Student Account Activation (Enrollment No. + Contact + OTP Verification)
 - Faculty / Staff Account Activation (Employee ID + Contact + OTP)
 - Secure Institutional Login (Student & Staff Portals)
 - Role Assignment (Institution-Provisioned, Backend-Authoritative)
 - Account Lifecycle Management (INVITED, ACTIVE, SUSPENDED, GRADUATED)
 - Credential Security (User-Owned Passwords, Bcrypt Hashing, Zero Plaintext)
 - Session Security (JWT, HttpOnly Cookies, Scoped RBAC Guard)
- 2. STUDENT ECOSYSTEM
 - Verified Campus Profile (Bio, Skills, GitHub, Portfolio)
 - Student Project Showcase (Details, Links, Tech Stack)
 - Explore Directory & Peer Project Discovery (Likes)
- 3. ACADEMIC MANAGEMENT
 - Timetable Schedule (Student & Faculty Views)
 - Attendance Tracking & Roll-Call Marking
 - Marks & Grade Book Evaluation
 - Assignments Publication & Submissions
- 4. INSTITUTIONAL COMMUNICATION
 - Official College Profile
 - Broadcast Notices & Bulletins
 - Campus Events Directory
- 5. CAMPUS AI ENGINE
 - Natural Language Campus Query Assistant
 - MongoDB Structured Data Router
 - Vector Search / RAG Policy Retriever
 - Anti-Hallucination Fallback Engine
- 6. PHASE 2 EXPANSION (PLANNED)
 - Curated Internships Portal
 - Hackathons & Team Matcher
 - Campus Community Discussion Feed
 - Lost & Found Digital Recovery System

22. Master Registry of Locked Decisions in Step 1 **LOCKED**

DECISION AREA	STATUS	FORMAL BASELINE DETERMINATION
Project Identity	LOCKED	Working name established as College OS ; final branding deferred.
College-Controlled Identity	LOCKED V1.1	Users do not choose their role; the institution provisions and authorizes all student, faculty, and administrative identities.
No Open Registration	LOCKED V1.1	Public self-registration with arbitrary role selection is strictly rejected across all user levels.
Student Account Activation	LOCKED V1.1	Students activate pre-provisioned records using Enrollment Number + Official Email/Phone + OTP; official academic identity is read-only.
Multi-Factor Security Rule	LOCKED V1.1	Enrollment Number alone is insufficient for activation; verified institution-controlled contact + OTP is mandatory.
User Password Ownership	LOCKED V1.1	Users create their own passwords during activation; stored via bcrypt; college staff does not maintain or see plaintext passwords.
Faculty Account Activation	LOCKED V1.1	Faculty activate provisioned records using Employee ID + Official Contact + OTP; permissions derive from institutional course mapping.
HOD / Principal Provisioning	LOCKED V1.1	Administrative accounts are provisioned exclusively through authorized institutional channels; zero public self-elevation.
Authoritative Role Backend	LOCKED V1.1	Roles are non-selectable and enforced by backend database authority; frontend role claims are never trusted.
Identity vs. Profile Boundary	LOCKED V1.1	Institution-controlled academic fields (immutable by user) are strictly separated from user-controlled showcase portfolio fields.
Account Lifecycle Model	LOCKED V1.1	Formal lifecycle states enforced: INVITED/NOT_ACTIVATED → ACTIVE → SUSPENDED → GRADUATED/INACTIVE.
Core Vision	LOCKED	Centralized digital campus ecosystem integrating academics, identity, discovery, and AI assistance.
Problem Definition	LOCKED	Information fragmentation across ERPs, WhatsApp, PDFs, and notices; lack of student identity.
Student Identity	LOCKED	Students maintain verified campus profiles showcasing bio, skills, GitHub, website, and projects.
Project Discovery	LOCKED	Centralized Explore directory to discover peers and projects; includes project liking.
Faculty Capability	LOCKED	Elevated permissions strictly bounded by admin-assigned classes and subjects.
Institutional Presence	LOCKED	Dedicated Official College Profile distinct from personal accounts; managed by authorized admins.
Core MVP Scope	LOCKED	11 foundational modules strictly defined; scope boundary protected from creep.
Phase 2 Modules	LOCKED	Internships, Hackathons, Campus Feed, and Lost & Found validated as Phase 2 additions.
AI Campus Assistant	LOCKED	Context-grounded assistant; college data is primary truth; zero speculative hallucinations.
Scope Philosophy	LOCKED	<i>Core First</i> → <i>Stabilize</i> → <i>Expand</i> lifecycle execution.

23. Decisions Intentionally Left for Subsequent Steps DEFERRED ROADMAP

STEP #	ROADMAP STAGE	SCOPE OF DECISIONS TO BE FORMALLY LOCKED
Step 2	User Types & Permission Matrix	Granular CRUD access control, role hierarchy (HOD, Faculty, Admin, Student), assigned-class access logic.
Step 3	Complete User Flows & Wireframes	Step-by-step UX interaction journeys, dashboard layout structures, state transitions.
Step 4	Feature & Module Breakdown	Detailed feature functional specifications, input validation rules, operational edges.
Step 5	AI + RAG Architecture	Embedding model selection, vector indexing, chunking strategies, prompt templates, fallback mechanics.
Step 6	Database & System Architecture	Formal MongoDB schemas, indexing strategies, API route specifications, caching layers.
Step 7	Technology Stack Finalization	Locking Next.js 14, Tailwind CSS, Node.js, Express, MongoDB Atlas, Mongoose, and LLM provider.
Step 8	Development Roadmap	Sprint plans, milestone deliverables, task assignments, test suite schedules.
Step 9	Technical Documentation	Software Requirements Specification (SRS), System Design Document (SDD), API docs.
Step 10	Project Presentation (PPT)	Executive slide decks, architecture diagrams, faculty defense presentations.
Step 11	Viva & FAQ Preparation	Defense questions, architectural justification, performance trade-off analysis.

24. The Source-of-Truth Feature Evaluation Rule LOCKED GOVERNANCE

THE 4-QUESTION FEATURE VALIDATION GATE

Before any new feature, field, or capability is introduced into the College OS codebase, it must affirmatively satisfy at least one of these four criteria:

1. **Does it directly resolve an identified, validated campus problem?**
2. **Does it reinforce the core vision of a unified, student-centric digital campus?**
3. **Does it deliver demonstrable, tangible value to a defined user role?**
4. **Does it strictly belong in the current development phase (Core MVP)?**

If the answer to all four is negative, the feature is immediately rejected or archived in the Phase 2 backlog.

25. Final Step 1 Summary & Sign-Off Status LOCKED

STEP 1 FOUNDATION SIGN-OFF SUMMARY

- **Platform Identity:** College OS — Centralized Digital Campus Platform.
- **Core User Roles:** Student, Faculty, Official Institution / Admin.
- **Core MVP Foundation:** 11 fully validated, interlinked modules.
- **Future Horizons:** 4 planned expansion modules reserved for Phase 2.
- **Governing Philosophy:** *Core First* → *Stabilize* → *Expand*.
- **Milestone Status:** ● FOUNDATION UPDATED & LOCKED • AUTHENTICATION & INSTITUTIONAL ONBOARDING LOCKED (v1.1).

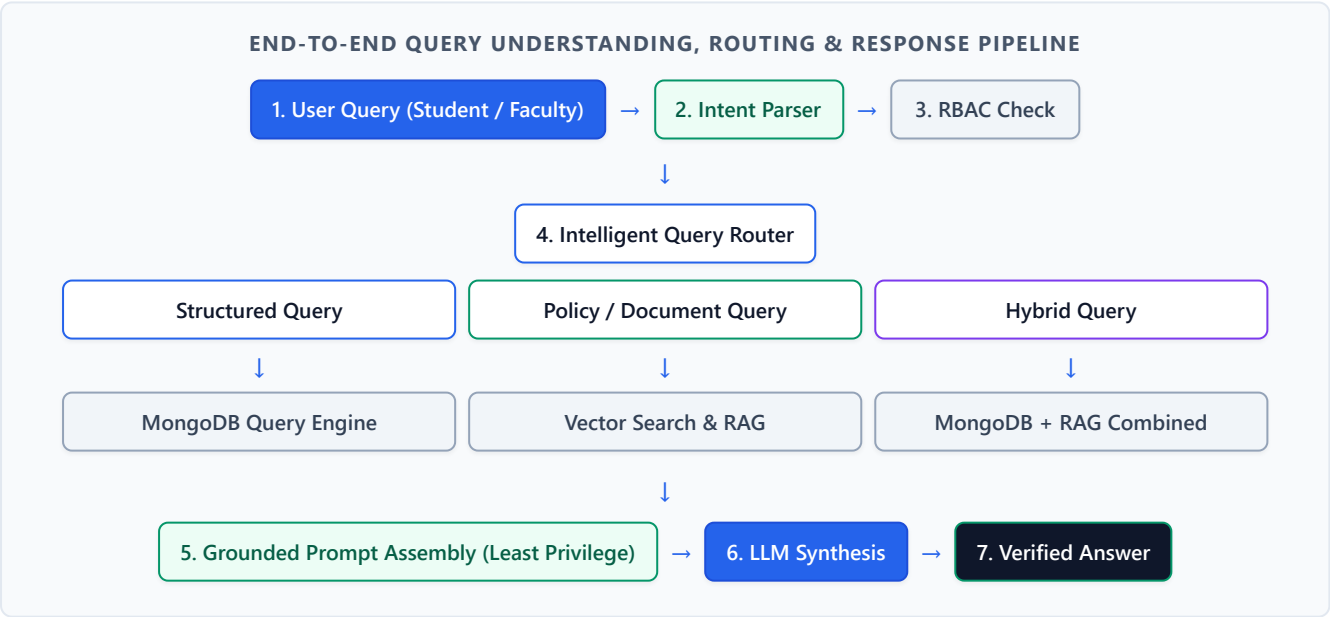
APPENDIX A • CAMPUS AI & RAG ARCHITECTURAL SPECIFICATION

Intelligent Campus Knowledge Engine • Architectural Blueprint

CORE AI ARCHITECTURAL PRINCIPLES

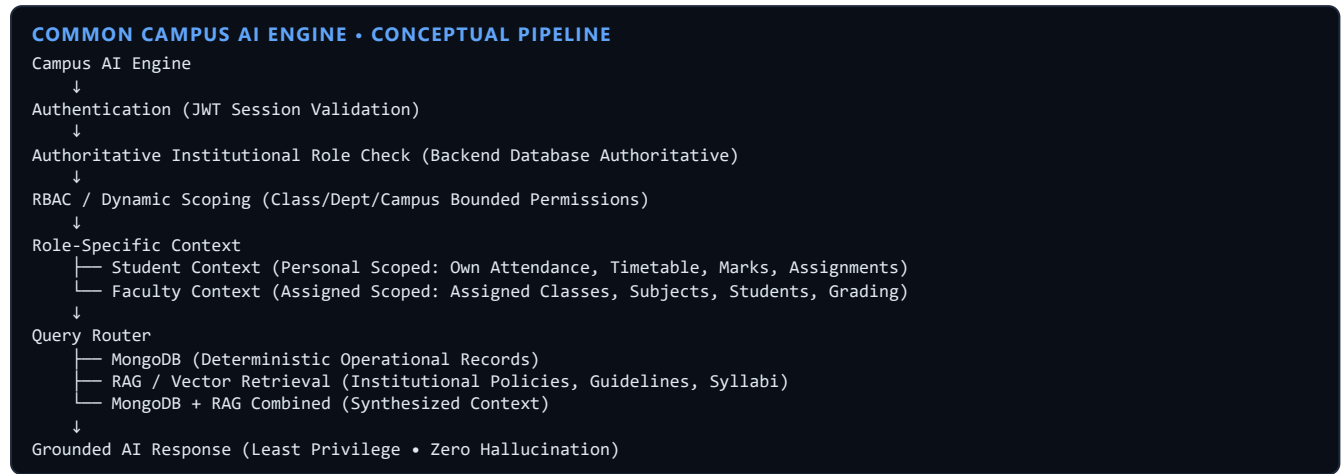
- **Not Generic ChatGPT:** Grounded strictly in authenticated college data.
- **College OS is the Primary Source of Truth:** Operates under strict domain boundaries.
- **One Common Campus AI Engine Across Roles:** Student AI and Faculty AI are not separate engines. One common intelligence engine operates across the institution; the authenticated user's role and RBAC permissions govern accessible context, MongoDB operational datasets, RAG document scope, suggested prompts, and available actions.
- **Structured Data → MongoDB Query Engine:** Real-time attendance, timetables, and marks are queried directly via deterministic database queries.
- **Unstructured Policies → RAG Vector Retrieval:** Institutional circulars, examination rules, and syllabi are retrieved via semantic vector search.
- **Hybrid Queries → MongoDB + RAG:** Complex inquiries combine both data streams into a synthesized context prompt.
- **Zero Hallucination Guarantee:** When required institutional records are absent, the engine returns an explicit "Information unavailable" response.
- **Cryptographic Security & Authorization:** Passwords, OTPs, session tokens, and secrets are strictly excluded from AI prompts; queries strictly enforce role authorization boundaries (RBAC).
- **Least-Privilege Context Assembly:** Only the absolute minimum necessary fields are retrieved and injected into the LLM context.
- **Vector/RAG is a Retrieval Layer:** MongoDB remains the permanent operational store; Vector Search (MongoDB Atlas Vector Search) acts as an indexing index.

A.1 Campus AI Query Processing Flow Pipeline



A.2 Role-Aware Context, Authorization & Faculty-Specific AI Experience

Campus AI is engineered around an essential architectural foundation: **Student AI and Faculty AI are NOT two separate AI engines**. There is **ONE common Campus AI engine** powering the entire institution. The authenticated user's role and RBAC permissions dynamically determine accessible context, data retrieval, suggested prompts, and actionable capabilities.



CONTEXT DIMENSION	STUDENT AI CONTEXT (PERSONAL SCOPE)	FACULTY AI CONTEXT (ASSIGNED SCOPE)
Authorized Information Scope	Students interact with Campus AI for information they are authorized to access: - Their attendance records - Their timetable (class/section schedule) - Their assignments, deadlines, and submissions - Their academic information, marks, and CGPA - Relevant notices and campus events - College policies and academic guidelines - Other permitted college and directory information	Faculty interact with the same Campus AI using a faculty-specific authorized context: - Assigned classes and assigned subjects - Assigned students in those classes/subjects - Attendance of assigned classes and students - Assignments and submissions for assigned classes - Marks and assessment results they are authorized to access - Faculty timetable and teaching schedules - Relevant notices and departmental announcements - Teaching resources and documents available to them - Relevant college policies and examination guidelines - Faculty-related tasks and academic workflows
Query Resolution Examples	Show my attendance. → Retrieves only that authenticated student's attendance records.	Show students below 75% attendance in my assigned CSE-302 class. → Retrieves attendance records strictly for students in classes assigned to that faculty member.
Unauthorized Request Enforcement	Show marks of Roll No. 42. → Denied: System rejects inquiry; unauthorized peer records are never exposed.	Show me private records of another faculty member. → Denied: System denies request; returns an appropriate unavailable-or-unauthorized response.
UI / Route Presentation	Campus AI prompt widget on the Student Dashboard and dedicated assistant interface with student-centric quick prompts.	Role-tailored Campus AI interface within the faculty portal (e.g., <code>/faculty/ai</code> or faculty dashboard assistant) with faculty-specific prompts and action scopes.

STRICT RBAC, LEAST-PRIVILEGE & GROUNDING GUARDRAILS

- Least-Privilege Context Assembly:** Every AI query must respect RBAC and least-privilege access. The AI must **NEVER expose**: unauthorized student data, other faculty's restricted data, private administrative information, passwords, credentials, session tokens, or any data the authenticated user does not have explicit permission to access.
- MongoDB & RAG Division of Responsibility:** MongoDB Atlas remains the application source of truth for structured transactional data. RAG / Vector Retrieval remains dedicated to policies, guidelines, circulars, and handbooks. Hybrid queries execute MongoDB + RAG in tandem. Role-based authorization must always apply *before* returning retrieved information.
- Zero Hallucination Mandate:** If authorized college information is unavailable in verified records, the AI must clearly state that the information is unavailable instead of hallucinating. Genuinely general, non-college queries may utilize general AI knowledge only in accordance with existing SOT rules.

APPENDIX B • MONGODB DATABASE ARCHITECTURE & SCHEMAS

Primary Operational Data Store • Production BSON Schemas

Storage Paradigm: MongoDB Atlas serves as the authoritative, transactional source of truth. Monospace schema representations below depict the exact field keys, BSON types, and relational foreign references across the 11 Core MVP modules.

Collection: usersRef by: Profiles, Notices

```
{
  _id: ObjectId,
  institutionId: ObjectId, // Ref -> collegeProfile._id
  email: String,           // Unique official
                           // institutional email
  passwordHash: String,    // Bcrypt hashed credential
                           // (user-created)
  role: String,            //
  "student"|"faculty"|"hod"|"principal"|"vp"|"admin"
  (Institution-assigned)
  accountStatus: String,   //
  "INVITED"|"ACTIVE"|"SUSPENDED"|"GRADUATED"
  isActive: Boolean,       // True once activation
  completed
  activatedAt: Date,        // Timestamp of activation
  lastLoginAt: Date,
  createdAt: Date,
  updatedAt: Date
}
```

Collection: studentProfilesRef: users._id, projects

```
{
  _id: ObjectId,
  userId: ObjectId,        // Reference -> users._id
  // --- INSTITUTION-CONTROLLED FIELDS (Immutable by
  // Student) ---
  studentId: String,       // Enrollment Number /
  University PRN
  rollNumber: String,      // Class Roll Number (where
  applicable)
  fullName: String,        // Official student name
  from college record
  officialEmail: String,   // Institution-issued email
  address
  phone: String,           // Institution-verified
  contact number
  department: String,      // Academic Branch (e.g.,
  "CSE")
  semester: Number,        // Current Semester (e.g.,
  7)
  section: String,         // Class Section (e.g., "A")
  academicStatus: String,  // "REGULAR" | "PROBATION" |
  "GRADUATED"
  // --- STUDENT-CONTROLLED PROFILE FIELDS
  (Customizable) ---
  bio: String,             // Student personal
  statement
  skills: [String],        // Verified technical skills
  (e.g., ["React", "MongoDB"])
  githubUrl: String,       // GitHub developer profile
  websiteUrl: String,      // Personal portfolio URL
  avatarUrl: String,       // Profile display image
  createdAt: Date,
  updatedAt: Date
}
```

Collection: facultyProfiles

Ref: users._id, assignments

```
{
  _id: ObjectId,
  userId: ObjectId,      // Reference -> users._id
  // --- INSTITUTION-CONTROLLED FIELDS (Immutable by
  // Faculty) ---
  facultyId: String,     // Official Employee /
  // Faculty ID
  fullName: String,      // Official faculty full
  // name
  officialEmail: String, // Institutional email
  // address
  phone: String,         // Institution contact
  // number
  department: String,    // Academic Department
  // (e.g., "CSE")
  designation: String,   // "Assistant Professor" |
  // "Professor"
  assignedClassIds: [String], // Classes dynamically
  // assigned by Admin
  assignedSubjectIds: [String], // Subjects dynamically
  // assigned by Admin
  academicStatus: String, // "ACTIVE" | "ON_LEAVE" |
  // "RELIEVED"
  // --- FACULTY-CONTROLLED PROFILE FIELDS ---
  cabinLocation: String, // Campus office location
  // (e.g., "Room 304, Tech Block")
  bio: String,           // Academic & research
  // overview
  avatarUrl: String,     // Profile display image
  createdAt: Date,
  updatedAt: Date
}
```

Collection: collegeProfile

Singleton / Institutional

```
{
  _id: ObjectId,
  institutionName: String, // "CSMU"
  institutionCode: String,
  establishedYear: Number,
  address: String,
  contactEmail: String,
  websiteUrl: String,
  authorizedAdminIds: [ObjectId], // Admins authorized
  institutionalBio: String,
  updatedAt: Date
}
```

Collection: projects

Ref: studentProfiles._id

```
{
  _id: ObjectId,
  studentId: ObjectId, // -> studentProfiles._id
  title: String,       // e.g., "College OS"
  tagline: String,
  description: String, // Markdown overview
  techStack: [String], // ["Next.js", "MongoDB"]
  githubUrl: String,
  liveDemoUrl: String,
  screenshotUrls: [String],
  likesCount: Number,  // Counter cache
  createdAt: Date,
  updatedAt: Date
}
```

Collection: projectLikes

Unique: { projectId, studentId }

```
{
  _id: ObjectId,
  projectId: ObjectId, // -> projects._id
  studentId: ObjectId, // -> studentProfiles._id
  createdAt: Date
}
```

Collection: timetables

Ref: facultyProfiles._id

```
{
  _id: ObjectId,
  department: String,
  semester: Number,
  section: String,
  academicYear: String, // "2025-2026"
  dayOfWeek: String,    // "Monday" ... "Friday"
  slots: [
    {
      startTime: String, // "09:00 AM"
      endTime: String,   // "10:00 AM"
      subjectCode: String,
      subjectName: String,
      facultyId: ObjectId, // -> facultyProfiles._id
      classroom: String
    }
  ],
  updatedAt: Date
}
```

Collection: attendance

Ref: studentProfiles, facultyProfiles

```
{
  _id: ObjectId,
  studentId: ObjectId, // -> studentProfiles._id
  classId: String,     // e.g., "CSE-7A"
  subjectId: String,   // e.g., "CS-701"
  date: Date,
  status: String,      // "present"|"absent"|"late"
  markedByFacultyId: ObjectId, // -> facultyProfiles
  remarks: String,
  createdAt: Date
}
```

Collection: marks

Ref: studentProfiles, facultyProfiles

```
{
  _id: ObjectId,
  studentId: ObjectId, // -> studentProfiles._id
  subjectId: String,
  semester: Number,
  examType: String,    // "internal_1" | "final"
  maxMarks: Number,
  marksObtained: Number,
  enteredByFacultyId: ObjectId, // -> facultyProfiles
  verifiedStatus: Boolean,
  updatedAt: Date
}
```

Collection: assignments

Ref: facultyProfiles, studentProfiles

```
{
  _id: ObjectId,
  title: String,
  description: String,
  subjectId: String,
  classId: String,
  assignedByFacultyId: ObjectId, // -> facultyProfiles
  dueDate: Date,
  maxScore: Number,
  attachmentUrls: [String],
  submissions: [
    {
      studentId: ObjectId,
      submittedAt: Date,
      fileUrl: String,
      score: Number
    }
  ],
  createdAt: Date
}
```

Collection: notices

Ref: users._id, collegeProfile

```
{
  _id: ObjectId,
  title: String,
  content: String,      // Markdown content
  publishedByRole: String, // "college"|"hod"|"faculty"
  authorId: ObjectId,   // -> users._id
  targetAudience: String, // "all"|"students"|"faculty"
  department: String,
  priority: String,     // "normal"|"urgent"
  attachmentUrls: [String],
  createdAt: Date
}
```

Collection: events

Campus Events & Symposiums

```
{
  _id: ObjectId,
  title: String,
  description: String,
  eventType: String,    // "academic"|"technical"
  organizer: String,
  venue: String,
  startDateTime: Date,
  endDateTime: Date,
  registrationUrl: String,
  bannerImageUrl: String,
  createdAt: Date
}
```

APPENDIX C • ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

Granular CRUD Authorization • Dynamic Class Assignment Logic

DYNAMIC FACULTY SCOPING RULE

A faculty member's authorization boundaries are dynamically bounded: they can ONLY inspect and manipulate academic data (attendance, marks, assignments) for classes and subjects explicitly assigned to them by administrative authority. They possess ZERO access to unrelated departments or classes.

Platform Module	Student Permissions	Faculty Permissions	Admin Permissions
Authentication	Activate Account (Enrollment + Contact + OTP), Login, Self-service Password Change	Activate Account (Employee ID + Contact + OTP), Login, Self-service Password Change	Provision Users (Import/Create Student & Staff records), Assign Roles, Revoke Access, Audit Lifecycle State
Student Profiles	Read All Directory, Update Student-Controlled Fields (Bio, Skills, GitHub, Portfolio). Zero edit access to Institution-Controlled Identity.	Read All Students in Assigned Classes/Subjects	Read All, Administrative Provisioning, Official Data Updates, Suspend Accounts
Projects & Likes	Create, Update Own, Delete Own, Like	Read All Projects, Inspect Details	Read All, Moderate / Remove Violations
Timetable	Read Personal Class Timetable	Read Assigned Teaching Schedule	Create, Update, Delete All Timetables
Attendance	Read Own Attendance Records Only	Create, Update Attendance for Assigned Classes	Read All, Audit, System Overrides
Marks	Read Own Academic Marks Only	Create, Update Marks for Assigned Subjects	Read All, Departmental Audit, Grade Locks
Assignments	Read Assigned, Submit Own Files	Create, Update, Delete for Assigned Classes; Grade	Read All, Administrative Auditing
Notices	Read Permitted Notices	Create Class/Dept Notices; Read All	Publish College-Wide Notices; Delete Any
Events	Read Events, Register	Read Events, Post Dept Events (if authorized)	Create, Update, Delete All Events
Campus AI	Natural Language Query (Scoped to Own Data)	Natural Language Query (Scoped to Assigned Data)	System Metrics, Retrieval Audits

OFFICIAL SOURCE OF TRUTH DOCUMENT SIGN-OFF

College OS • Final-Year Major Project • Department of Computer Science & Engineering • CSMU

PREPARED BY

Hamid Rza & Team

Lead Developer / 7th Sem

REVIEWED BY

[Project Guide Name]

Faculty Project Guide

STATUS

 BASELINE LOCKED

Source of Truth v1.1